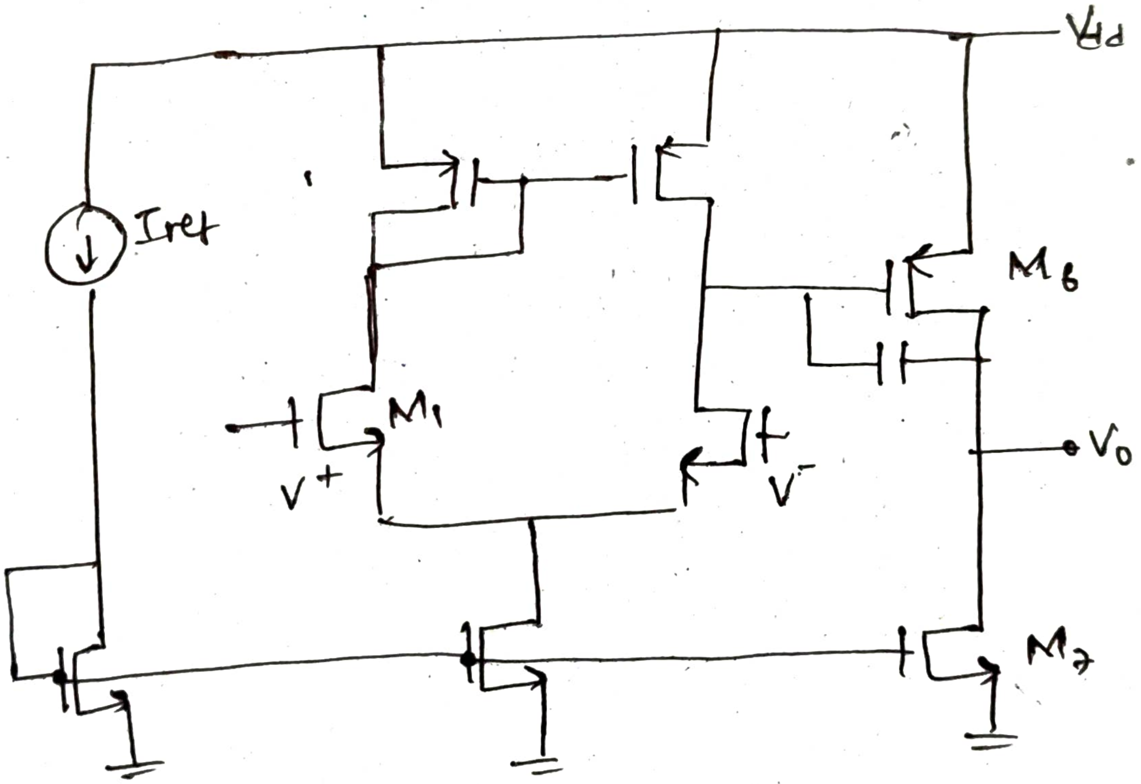
2-Stage OTA

Specifications:-

DC Gain > 60dB
GBW = 5 MHz
Process = 180nm
PM ≥ 60°
Slew Rate ≥ 10 V/μsec
ICMR (+) = 1.6 V
ICMR (-) = 0.8 V

$C_L = 10$ pF
$V_{dd} = 1.8$ V
$\mu_n C_{ox} = 246.74\ \mu A/V^2$
$\mu_p C_{ox} = 92.75\ \mu A/V^2$

Circuit diagram:-



Method of Design:-

→ $\frac{W}{L}$ of M_3, M_4 is found using ICMR(+)
→ $\frac{W}{L}$ of M_1, M_2 is found using GBW
→ I_5 is found using slew rate
→ $\frac{W}{L}$ of M_5 found using ICMR(-)

→ $\frac{W}{L}$ of M_6 using phase margin and zero location

→ $\frac{W}{L}$ of M_7 by M_5 (related)

→ C_c using C_L and phase margin.

Calculations:-

① For PM > 60° : $C_c \geq 0.22\, C_L$

As: $$\Phi_M = 60^\circ = 180^\circ - \tan^{-1}\left(\frac{\omega}{|P_1|}\right) - \tan^{-1}\left(\frac{\omega}{|P_2|}\right) - \tan^{-1}\left(\frac{\omega}{z_1}\right)$$

@ GBW (since GBW = Open loop UGF) (f = 1)

$$120^\circ = \tan^{-1}\left(\frac{GB}{|P_1|}\right) + \tan^{-1}\left(\frac{GB}{|P_2|}\right) + \tan^{-1}\left(\frac{GB}{z_1}\right)$$

$$= \tan^{-1}(\text{DC gain}) + \tan^{-1}\left(\frac{GB}{|P_2|}\right) + \tan^{-1}(0.1)$$

$[z_1 = 10GB]$

$A_V(0) \rightarrow \infty$

$$24.3^\circ = \tan^{-1}\left(\frac{GB}{|P_2|}\right)$$

$$|P_2| \geq 2.2\, GB$$

$(z_1 \geq 10GB)$ $\frac{g_{m6}}{C_C} \geq 10\frac{g_{m2}}{C_C} \Rightarrow g_{m6} \geq 10\, g_{m2}$

$$\frac{g_{m6}}{C_2} \geq 2.2\, \frac{g_{m2}}{C_C} \Rightarrow C_c \geq 0.22\, C_L$$

→ $C_L = 10\,pF$

$C_c \geq 2.2\,pF$

Let $\boxed{C_c = 3\,pF}$

(2) I_5 : Slew rate $= \frac{I_5}{C_c}$

$I_5 = 10\,V/\mu s \times 3\,pF = 30\,\mu A$

(3) <u>M_1 & M_2</u>

$gm_1 = GBW \cdot C_c \cdot 2\pi = 94.24\,\mu S$

$$\left(\frac{W}{L}\right)_1 = \frac{gm^2}{\mu_n C_{ox} \times 2 I_D} = \frac{(94.24\mu)^2}{246.79 \times 30 \times \mu^2} = 1.2$$

(4) <u>M_3 & M_4</u>

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2 I_{D3}}{\mu_p C_{ox} \left[V_{DD} - ICMR(+) - V_{t3\,max} + V_{t1\,min}\right]^2}$$

$V_{t3\,max} = 0.433\,V$

$V_{t1\,min} = 0.31\,V$

$$\left(\frac{W}{L}\right)_3 = \frac{30}{92.75\,(1.8 - 1.6 - 0.43 + 0.31)^2} = 54.55 \approx 55$$

$$\left(\frac{W}{L}\right)_{3,4} = 55$$

(5) For M5, M8

$V_{D sat 5} = ICMR(-) - V$

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since $ICMR(+) = V_{DD} - \sqrt{\frac{I_5}{\beta_3}} - V_{t03\,max} + V_{t1\,min}$

as $ICMR(+) - V_{t1\,min} < V_{G3}$

$V_{G3} = V_{DD} - V_{ov3} - V_{tp}$

(5) For M5 & M8

$$V_{D sat 5} = ICMR(-) - V_{SS} - \sqrt{\frac{I_5}{\beta_1}} - V_{th1\,max}$$

since $ICMR(-) - V_{ov1} - V_{th1\,max} = V_{dsat5}$

$$V_{D sat5} = 0.8 - 0 - 0.319 - 0.38 = 0.1\,V$$

$$\left(\frac{W}{L}\right)_5 = \frac{2 I_5}{\mu_n C_{ox} (V_{D sat 5})^2} = \frac{2 \times 30}{246.74 \times 0.1^2} = 24.31$$

$$\left(\frac{W}{L}\right)_{5,8} = 24$$

(6) For M6

$$g_{m6} = 10 g_{m1} = 942.4\,\mu$$

$$g_{m4}^2 = 2 I_4 \mu_p C_{ox} \times \left(\frac{W}{L}\right)_4 \Rightarrow g_{m4} = \sqrt{30 \times 92.75 \times 55}$$

$$g_{m4} = 391.20\mu$$

$$\left(\frac{W}{L}\right)_6 = \frac{g_{m6} \times \left(\frac{W}{L}\right)_4}{g_{m4}}$$ [Since M4 & M6 have equal overdrives]

$= 132.49 \approx 132$

(7) For M_7:

$$I_6 = \frac{\left(\frac{W}{L}\right)_6 * I_4}{\left(\frac{W}{L}\right)_4} = \frac{132 * 15\mu}{55} = 36\mu A$$

$$\left(\frac{W}{L}\right)_7 = \frac{I_7 * \left(\frac{W}{L}\right)_5}{I_5} = 28$$

Here for no systematic offset:

$$2 * \frac{\left(\frac{W}{L}\right)_7}{\left(\frac{W}{L}\right)_5} = \frac{\left(\frac{W}{L}\right)_6}{\left(\frac{W}{L}\right)_4}$$ is satisfied

Data:-

$I_{dc} = 30\mu A$ $C_c = 3pF$ $C_L = 10pF$ Let $L = 500nm$

$M_{1,2}$	$W = 600nm$	$L = 500nm$
$M_{3,4}$	$W = 27\mu m$	$L = 500nm$
M_5	$W = 12\mu m$	$L = 500nm$
M_6	$W = 66\mu m$	$L = 500nm$
M_7	$W = 14\mu m$	$L = 500nm$
M_8	$W = 12\mu m$	$L = 500nm$

Analysis:-

Total Open loop gain : 65dB ≥ 60dB

GBW = 3dB of closed loop : 5.07 MHz

Phase margin = 63°

Phase margin was expected to be 60°.